

AKASH KUNDU

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Portfolio: akashkundu114.github.io | GitHub: github.com/AkashKundu114 | LinkedIn: linkedin.com/in/akashkundu114

PROFESSIONAL SUMMARY

Software Development Engineer with strong foundations in **Data Structures & Algorithms (DSA)**, **Object-Oriented Programming (OOP)**, and **Distributed System Design**. Experienced in architecting low-latency microservices (~**9,856 QPS** dispatch, **<0.10 ms** latency), accelerating model serving runtimes by **2.15x** via ONNX graph compilation, and engineering secure, multi-tenant databases with PostgreSQL Row-Level Security, asynchronous workers, and automated CI/CD pipelines.

TECHNICAL SKILLS

Languages: Python, Java, C++, JavaScript, TypeScript, SQL (PostgreSQL, SQLite), HTML/CSS

Core CS Concepts: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), System Design, REST APIs, Microservices, Concurrency, Asynchronous Programming

Backend & Frameworks: FastAPI, React 19, Node.js/Electron, WebSockets, Asyncio, Celery, Redis (Caching & Rate Limiting), PyTorch, OpenCV

Databases & Storage: PostgreSQL (Row-Level Security, pgvector), SQLite (WAL Mode), ChromaDB, MongoDB

Tools, Cloud & DevOps: Git, Docker, Kubernetes, Linux/Unix CLI, Microsoft Azure, CI/CD (GitHub Actions), Postman, Pytest (TDD), Vitest, Playwright

WORK EXPERIENCE

Purposive AI | *Software Engineering Intern (Backend & Systems)*

Kolkata, India (Hybrid) | Jul 2026 - Present

Technologies: Python, FastAPI, React, PostgreSQL, Celery, Sarvam AI, Ollama, Azure Container Apps, Pytest

- Architected an asynchronous ledger processing pipeline using FastAPI, React, and Celery, converting conversational Bengali audio into verified, bank-submittable PDF records and cutting manual bookkeeping effort by **80%**.
- Engineered an automated Sarvam AI to local Ollama fallback cascade circuit breaker, eliminating single points of failure, preventing vendor lock-in, and slashing cloud inference costs by **87%**.
- Designed asynchronous task worker queues with PostgreSQL pgvector state tracking, maintaining **sub-500ms** API turnaround and sustaining **99.9% uptime** against cloud production SLAs.
- Enforced Test-Driven Development (TDD) by authoring a **156-test** Pytest test suite, achieving a **100% pass rate** and eliminating regression defects across financial calculation and citation grounding logic.
- Implemented token-bucket API rate limiting, structured transactional logging, and strict input sanitization to guarantee complete auditability across concurrent user transactions.

PROJECTS

C.O.P.P.E.R. - Local AI Operating System & Microservices Engine

github.com/AkashKundu114/COPPER

Technologies: Electron, React 19, FastAPI, Python, Ollama, Redis, SQLite (WAL), ChromaDB, Docker, Pytest, Vitest

- Architected a 30-agent orchestration hierarchy with a Topological Failure-Predicting (TFP) cascade router executing dispatches at ~**9,856 QPS** (**<0.10 ms** cache dispatch, ~**2,050 QPS** full combinatorial) across 1,390 benchmark tests.
- Engineered the Forge Sandbox with AST analysis, 3-stage self-healing retry loops, and DFM-Guard 4-tier friction protocol, achieving **100% threat catch sensitivity across 350 adversarial triggers**.
- Developed long-term Bayesian epistemic memory backed by SQLite Write-Ahead Logging (WAL) and ChromaDB, coupled with an in-line Zero-Trust data firewall scrubbing PII and credentials prior to persistence.
- Shipped an Electron desktop app (**<1.0 GB RAM** footprint) in React 19 featuring IPC bridge communication, real-time hardware telemetry monitoring, and **538 passing automated tests** (500 Pytest + 38 Vitest).

OphthalmolAI - Telemedicine Screening Backend & ML Platform

github.com/AkashKundu114/OphthalmolAI

Technologies: FastAPI, Python, PyTorch, ONNX Runtime, PostgreSQL, React 19, Docker, OpenCV, Pytest | Live: ophthalmo-ai-mu.vercel.app

- Architected a calibrated tri-backbone vision ensemble (DenseNet-201 + ConvNeXt + EfficientNet) with Platt temperature scaling (**0.0644 ECE**), achieving **85.18% test accuracy** and **0.9805 Macro AUROC** across 938 held-out clinical fundus images.
- Engineered pre-inference optical domain guardrails (OAC-DG) evaluating aperture geometry and chromophores to deterministically intercept non-fundus imagery with HTTP 422 before GPU allocation.
- Optimized production inference serving to **84.2 ms p50 latency (2.15x speedup)** at 17.3 QPS via ONNX Runtime FP16 graph compilation, complemented by a sub-50 ms in-browser edge screening pipeline.
- Implemented multi-tenant PostgreSQL Row-Level Security (RLS) with cryptographic tenant scoping (JWT claims) and role-based access control (RBAC), verified across **190 passing automated tests**.

EDUCATION

Techno India University | *Bachelor of Technology (B.Tech) in Computer Science & Engineering*

Kolkata, India | 2023 - 2027

• **Academic Standing:** CGPA: **7.5 / 10.0** | **Relevant Coursework:** Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, System Design

Arambagh Vivekananda Academy | *Higher Secondary Certificate (CBSE - Class XII Science)*

Arambagh, India | 2021 - 2023

• **Academic Score:** **71.6%** | Focus in Mathematics, Physics, and Computer Science

CERTIFICATIONS

• **Microsoft Certified:** Azure AI Developer Associate (AI-103)

• **Oracle Certified:** AI Database Foundations Associate (1Z0-1195)

• **Microsoft Certified:** ML Operations Engineer Associate (AI-300)

• **Oracle Certified:** Agentic AI Foundations Associate (1Z0-1157)